

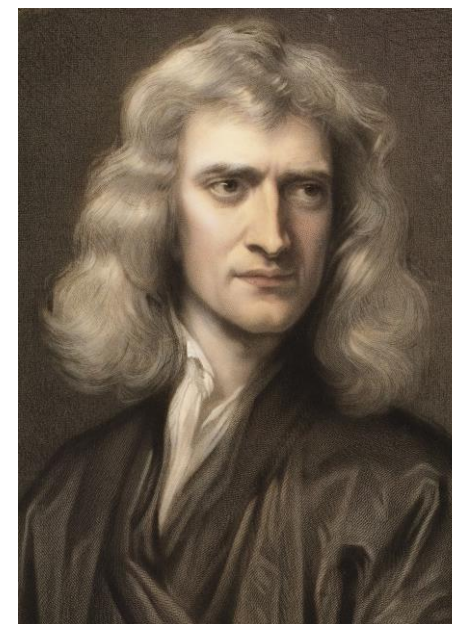
Math Primer

(7. Ordinary Differential Equations)

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Isaac Newton
(1643-1727)



G.W. Leibniz
(1646-1716)



Ordinary Differential Equations

- ▶ Often in physics or engineering, we arrive at a differential equation first and then seek for solutions to the equation. Quant finance is no different. Derivative pricing problems manifest as differential equations (Black Scholes famous PDE)
- ▶ An ordinary differential equation involves ordinary derivatives such as

$$x, \quad y, \quad \frac{dy}{dx}, \quad \frac{d^2y}{dx^2}, \quad \dots, \quad \frac{d^ny}{dx^n}$$

in an equation $F(x, y, y', y'', \dots, y^{(n)}) = 0$

- ▶ Order of the DE is the highest order of differentiation in the DE
- ▶ Degree of the DE is the power of the highest order of differentiation

Ordinary Differential Equations

► Examples

DE	Order	Degree
$y' = 3y$	1	1
$(y')^3 + 4 \ln y = x^3$	1	3
$(y^{(4)})^4 + x^2(y'')^5 + y = 0$	4	4
$y'' = \sqrt{y' + y + x}$	2	2
$2y'' + x(y')^3 - xy = 0$	2	1

Linear ODE



- ▶ A linear ODE of order n has
 - ▶ powers of all derivatives = 1
 - ▶ coefficients are only functions of x
 - ▶ only linear term of y is allowed (can be thought of as zeroth derivative)

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

Linear ODE



► Examples of linear ODE

DE	Linear ?
$2xy'' + x^2y' - (x + 3)y = x^2$	Yes
$yy'' + xy' + y = 10$	No
$y'' + \sqrt{y'} + y = x^2$	No
$y^{(4)} + y^4 = 0$	No

Solving a DE

- ▶ We can either solve y in general form with arbitrary constants or solve more explicitly if more conditions are given

- ▶ Exercise : Find the DE by eliminating the constants
 - a) $y = x^3 + ce^{-3x}$
 - b) $y = c_1e^{-x} + c_2e^{2x}$

- ▶ Exercise : Show that y is a solution to the DE
 - a) $y = c_1 \sin 2x + c_2 \cos 2x$, DE : $y'' + 4y = 0$

- ▶ Conversely, an n th order ODE will have a solution with n arbitrary constants

IVP vs BVP

- ▶ We can assign specific values to the constants $(c_1, c_2, \dots, c_n)$ if values of $(y, y', y'', \dots)$ etc. are given. Of course, one needs n such conditions to solve for n arbitrary constants.
- ▶ An **Initial Value Problem** (IVP) is the setup where all values of y and its derivatives are given for the same x

Example : $y'' + 2y' = 5e^x$ and $y(2) = 1$, $y'(2) = 3$

- ▶ A **Boundary Value Problem** (BVP) is the setup where the values of y and its derivatives are given for different x

Example : $y'' + 2y' = 5e^x$ and $y(0) = 1$, $y(1) = 1$

IVP vs BVP



▶ Exercise : Which of the following functions is the solution to the IVP

a) $y_1 = \sin 2x$ b) $y_2 = x$ c) $y_3 = \frac{1}{2} \sin 2x$

$$\text{IVP : } y'' + 4y = 0 ; y(0) = 0 ; y'(0) = 1$$

First order ODE



► First order DE is of the form

$$y' = f(x, y)$$

Or could be of the form $F(x, y, y') = 0$ which can be re-organized as above

Example :

$$xy' + 2xy - y = 0$$

$$\Rightarrow y' = \frac{y - 2xy}{x}$$

First order ODE

► Type : **One Variable Missing**

y missing : $y' = f(x)$, solution is $y = \int f(x)dx$

x missing : $y' = f(y)$, solution is $x = \int \frac{1}{f(y)} dy$

Example :

a) $y' = y$, $y(0) = 1$

b) $y' = x^2$, $y(-1) = 0$

First order ODE



▶ Type : **Variable Separable Form**

$$y' = f(x)g(y)$$
$$\Rightarrow \int \frac{dy}{g(y)} = \int f(x)dx$$

Example :

$$y' = \frac{x^2 + 2}{y}$$

First order ODE



► Type : **Linear**

$$y' + P(x)y = Q(x)$$

To solve this, multiply Integrating Factor (I.F) = $e^{\int P dx}$ on both sides. LHS will become an exact differential of $y \cdot IF$

Example :

$$a) \quad y' - \frac{1}{x}y = x^2$$

$$b) \quad \frac{dx}{dy}(1 + ye^x) = e^x$$

Second order ODE



► Typically, second order ODE (degree 1) takes the form

$$y'' = f(x, y, y')$$

The solution will involve 2 arbitrary constants.

Second order ODE



▶ Case I : y' , y are missing

$$y'' = f(x)$$

Integrate y twice

Example :

$$y'' = 4x$$

Second order ODE



► Case II : y is missing

$$y'' = f(x, y')$$

Take $y' = t$, so the equation becomes $t' = f(x, t)$

Example :

$$xy'' + 2y' = x^3$$

Second order ODE



► Case III : y', x are missing

$$y'' = f(y)$$

Take $y' = t$, and then write $y'' = \frac{dt}{dx} = \frac{dt}{dy} \frac{dy}{dx} = t \frac{dt}{dy}$, so the equation becomes $t \frac{dt}{dy} = f(y)$ which is in variable separable form.

Example :

$$y^3 y'' = 4$$

Second order ODE



▶ Case IV : x is missing

$$y'' = f(y, y')$$

Take $y' = t$, and then write $y'' = \frac{dt}{dx} = \frac{dt}{dy} \frac{dy}{dx} = t \frac{dt}{dy}$, so the equation becomes $t \frac{dt}{dy} = f(y, t)$ which is a first order ODE.

Example :

$$y^2 y'' = 4y'$$

Linear ODEs in general

▶ General nth order Linear ODE is of the form

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \dots + a_1(x)y^{(1)} + a_0(x)y = g(x)$$

Using the operator $D := \frac{d}{dx}$; $D^2 := \frac{d^2}{dx^2}$; ... ; $D^k := \frac{d^k}{dx^k}$ and so $D^k y := \frac{d^k y}{dx^k}$

Now Define Linear Differential Operator

$$L = a_n(x)D^n + a_{n-1}(x)D^{n-1} + \dots + a_1(x)D + a_0(x)$$

Above DE now can be re-written as $Ly = g$

Linear ODEs



► Note that L is a linear operator because

$$L(y_1 + y_2) = L(y_1) + L(y_2)$$

$$L(cy) = cL(y)$$

► if $g(x) = 0$, the DE is called **Homogeneous**

The general solution to the DE is given by the sum of

- a) Solution to the homogeneous equation (complementary function y_c) , $Ly_c = 0$ and
- b) Solution to the original DE (particular integral) , $Ly_p = g$

If there are multiple complementary solutions $(y_1, y_2, \dots, y_n)$, then linear combination of them $(\lambda_1 y_1 + \lambda_2 y_2 + \dots + \lambda_n y_n)$ is also a solution

Linear ODE with constant coefficients

► Consider a homogeneous DE : $Ly = ay'' + by' + cy = 0$

Try a solution of the form $y = e^{\lambda x}$, then

$$Ly = (aD^2 + bD + c)e^{\lambda x}$$

We get

$$a\lambda^2 + b\lambda + c = 0 \quad (\text{Auxiliary Equation})$$

Case I : $b^2 - 4ac > 0$, $y = c_1 \exp(\lambda_1 x) + c_2 \exp(\lambda_2 x)$

Case II : $b^2 - 4ac = 0$, $y = (c_1 + c_2 x) \exp(\lambda x)$

Case III : $b^2 - 4ac < 0$, $y = e^{px}(A \cos qx + B \sin qx)$, where $p = -\frac{b}{2a}$, $q = \frac{1}{2a}(\sqrt{b^2 - 4ac})$

Linear ODE with constant coefficients



► Examples :

$$a) \quad y'' - 3y' - 4y = 0$$

$$b) \quad y'' - 8y' + 16y = 0$$

$$c) \quad y'' - 3y' + 4y = 0$$

General nth order ODE with const. coefficients

▶ General nth order ODE with constant coefficients

$$Ly = 0$$

$$L = a_n D^n + a_{n-1} D^{n-1} + \dots + a_1 D + a_0$$

Auxiliary Equation becomes

$$a_n \lambda^n + a_{n-1} \lambda^{n-1} + \dots + a_1 \lambda + a_0 = 0$$

Case I : n distinct real roots , $y = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x} + \dots + c_n e^{\lambda_n x}$

Case II : r repeated real roots , $y = e^{\lambda x} (c_1 + c_2 x + c_2 x^2 + \dots + c_r x^{r-1})$

Case III : r repeated complex roots of the form $p + iq$, then

$$y = e^{px} (\alpha_1 + \alpha_2 x + \alpha_2 x^2 + \dots + \alpha_r x^{r-1}) \cos qx + e^{px} (\beta_1 + \beta_2 x + \beta_2 x^2 + \dots + \beta_r x^{r-1}) \sin qx$$

General nth order ODE with const. coefficients



► Examples :

$$a) \quad y^{(4)} - 5y'' + 6y = 0$$

$$b) \quad y^{(6)} - 5y^{(4)} = 0$$

$$c) \quad y^{(4)} + 2y'' + y = 0$$



Non-homogeneous case – Particular Solution

► Note that the general solution $y = y_c + y_p$

The particular-solution (P.I.) can be attempted using Guess Work by assuming the same form as g

Case I – $g(x)$ is a polynomial i.e., $g(x) = p_0 + p_1x + \cdots + p_mx^m$

Case II – $g(x)$ is exponential i.e., $g(x) = Ce^{kx}$ (provided k is not a root of A.E.)

Case III – $g(x)$ is trigonometric e.g., $g(x) = \sin ax$ (provided ia is not a root of A.E.)

Case IV – $g(x)$ is a combination of above

Non-homogeneous case – Particular Solution

► Examples :

$$a) \quad y'' + 3y' + 2y = 3e^{5x}$$

$$b) \quad y'' + 3y' + 2y = x^2$$

$$c) \quad y'' - 5y' - 6y = \cos 3x$$

Linear ODE with Variable Coefficients – Euler Equation



▶ Cauchy Euler equation is given as

$$Ly = ax^2 \frac{d^2y}{dx^2} + bx \frac{dy}{dx} + cy = g(x)$$

Note that the coefficient has the same power of x as the order of the derivative

The equation is linear. To solve the homogeneous part $Ly = 0$, we try $y = x^\lambda$. Upon substitution, it yields the **A.E.** $a\lambda^2 + b\lambda + c = 0$

Case I : $b^2 - 4ac > 0$, $y = Ax^{\lambda_1} + Bx^{\lambda_2}$

Case II : $b^2 - 4ac = 0$, $y = (A + B \ln x) x^\lambda$

Case III : $b^2 - 4ac < 0$, $y = x^p (A \cos(q \ln x) + B \sin(q \ln x))$, where $p = -\frac{b}{2a}$, $q = \frac{1}{2a}(\sqrt{b^2 - 4ac})$

Linear ODE with Variable Coefficients – Euler Equation



► Examples :

$$a) \quad x^2 y'' - 2xy' - 4y = 0$$

$$b) \quad x^2 y'' - 7xy' + 16y = 0$$

$$c) \quad x^2 y'' - 3xy' + 13y = 0$$